NTIRE 2026 Remote Sensing Infrared Image Super-Resolution Challenge Factsheet
-Quality-Aware Hybrid Attention Transformer for Infrared Remote Sensing Image Super-Resolution-

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1. Team Details

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diffusion
Best Score:
54.436

2. Method Details

2.1. Method Overview

Our approach is built upon the Hybrid Attention Transformer (HAT) [1]. In the challenge dataset, the low-resolution (LR) images are generated from high-resolution (HR) images using hand-crafted interpolation. However, the HR images themselves exhibit significant quality variations. Specifically, some HR images correspond to true high-resolution satellite observations, while others originate from lower-resolution sources or contain noticeable degradations such as sensor noise or JPEG compression artifacts. As a result, the HR supervision is heterogeneous, and treating all training samples equally may introduce inconsistent learning signals.

To address this issue, we propose a Quality-Aware HAT (QAHAT) that explicitly models the quality variations of HR images. Our framework introduces two complementary components: a Global Quality Branch and a Local Quality Branch. The global branch estimates the overall degradation characteristics of the input, while the local branch captures spatially varying artifacts. By incorporating both global and local quality cues, the proposed method enables the backbone network to adapt its feature processing to different degradation conditions.

2.2. Global Quality Modeling

We introduce a lightweight Quality Estimation Module (QEM) to estimate the overall degradation of the input image. The QEM consists of several strided convolutional layers followed by global pooling and fully connected layers, producing a compact quality descriptor that summarizes the degradation characteristics of the input image. The estimated quality vector is injected into the backbone through multiple Source Quality Adapters (SQA) inserted in the deep feature extraction stages of HAT. Each SQA modulates intermediate feature maps using channel-wise scaling and bias parameters derived from the quality representation, allowing the network to dynamically adjust feature responses according to the predicted degradation level.

2.3. Local Artifact Modeling

In addition to global quality variations, infrared remote sensing images may contain spatially localized artifacts. To model these local degradations, we introduce an Local Quality-Aware Module (LQAM). The LQAM extracts artifact-aware features using a downsampling pathway that provides a larger receptive field for capturing contextual

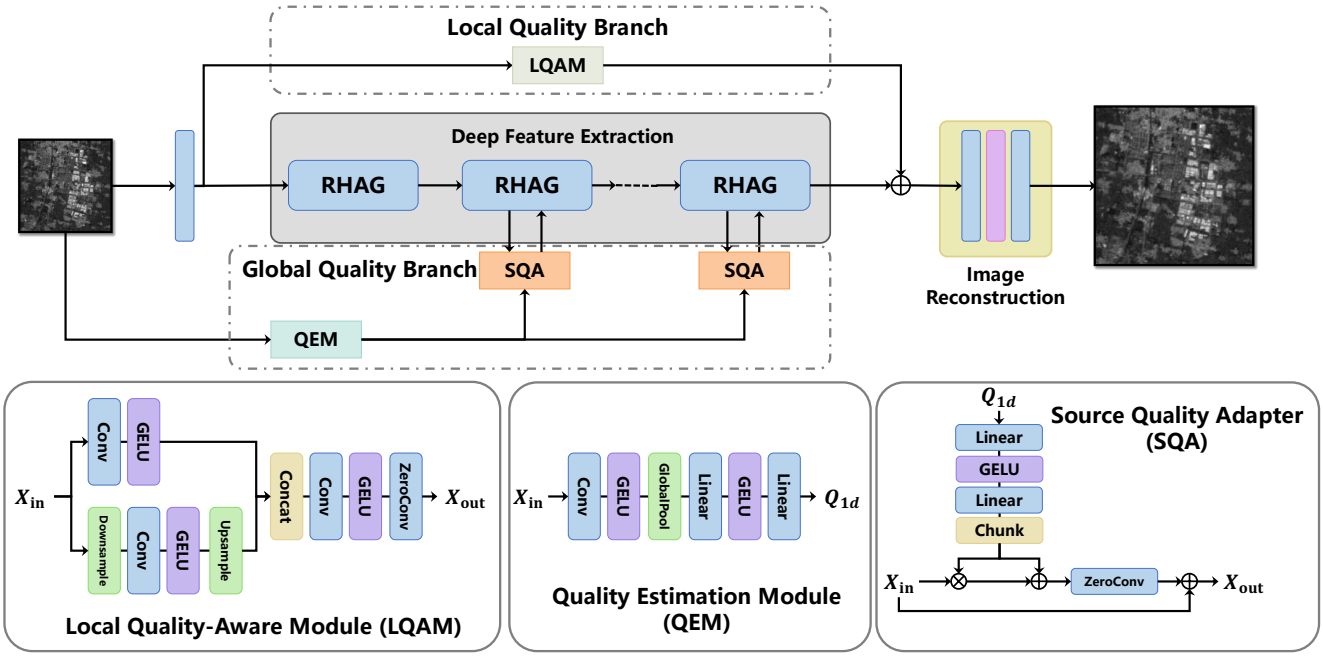

Figure 1. Architecture of the proposed Quality-Aware Hybrid Attention Transformer (QAHAT).

degradation patterns. The resulting artifact representation is fused with the backbone features to guide the reconstruction process in regions affected by degradations. Similar to the global branch, the fused features are integrated into the HAT backbone through a residual connection to ensure stable training.

By jointly modeling global degradation characteristics and local artifact patterns, the proposed QAHAT improves the robustness of the super-resolution process and produces more reliable reconstructions for infrared remote sensing images.

2.4. Implementation Details

2.4.1. Datasets

In addition to the training data provided by the challenge, we incorporate the publicly available thermal infrared remote sensing dataset SatVideoIRSDT [2] as supplementary training data. Since SatVideoIRSDT is a video dataset, we randomly sample one frame from each video sequence to construct an auxiliary thermal infrared image dataset. The sampled frames are then combined with the official competition training data to form the final training set.

2.4.2. Training Strategy

The proposed model is trained on image patches of size 256×256. We adopt the Adam optimizer with an initial learning rate of 5×10^{-5} and momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The learning rate is updated using a MultiStepLR scheduler, where the learning rate is reduced by a factor of 0.5 at 20k and 40k iterations. The model is trained for 200k iterations in total. An exponential moving average (EMA) with a decay factor of 0.999 is maintained to stabilize the training process. Following the evaluation protocol of the challenge, the training objective is designed to align with the official metric. The loss function is defined as $\mathcal{L} = -\mathrm{PSNR}(I_{\mathrm{sr}}, I_{\mathrm{gt}}) - 20 \cdot \mathrm{SSIM}(I_{\mathrm{sr}}, I_{\mathrm{gt}})$. The backbone network is initialized using pretrained weights of the HAT-L model, while the newly introduced modules are randomly initialized and jointly optimized during training.

2.4.3. Testing Strategy

During inference, we adopt a self-ensemble strategy by applying flip and rotation transformations to the input image. The outputs corresponding to different transformed inputs are inversely transformed and averaged to produce the final super-resolved result.

References

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